

Chosen data file. `project_obs_data_rady.txt`

Chosen estimator. *Joseph-stabilized sequential filter*

(a) Chosen estimator and governing equations

For this study, the chosen estimator is the **Joseph-stabilized sequential filter**. The state vector is taken as

$$\mathbf{X} = [\mathbf{r} \quad \dot{\mathbf{r}} \quad \mu \quad J_2 \quad C_D \quad \mathbf{x}_{s_i}]^T,$$

where $\mathbf{r}$ and $\dot{\mathbf{r}}$ are the ECI position and velocity of the satellite, μ is the Earth's gravitational parameter, J_2 is the oblateness coefficient, C_D is the drag coefficient, and $\mathbf{x}_{s_i}$ collects the solve-for ground-station coordinates (stations 337 and 394; station 101 is held fixed). The filter works with the deviation $\hat{\mathbf{x}}$ of the state from a reference trajectory $\mathbf{X}^*$, and the reference and the 9×9 state transition matrix Φ are propagated together with the two-body, J_2 , and drag dynamics.

The time-update equations used in the implementation are

$$\begin{aligned} \hat{\mathbf{x}}_k^- &= \Phi(t_k, t_{k-1}) \hat{\mathbf{x}}_{k-1}^+, \\ P_k^- &= \Phi(t_k, t_{k-1}) P_{k-1}^+ \Phi(t_k, t_{k-1})^T. \end{aligned}$$

The measurement-update equations are

$$\begin{aligned} \boldsymbol{\nu}_k &= \mathbf{Y}_k - G(\mathbf{X}_k^*, t_k), \\ K_k &= P_k^- \tilde{H}_k^T \left(\tilde{H}_k P_k^- \tilde{H}_k^T + R \right)^{-1}, \\ \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + K_k \left(\boldsymbol{\nu}_k - \tilde{H}_k \hat{\mathbf{x}}_k^- \right), \\ P_k^+ &= (I - K_k \tilde{H}_k) P_k^- (I - K_k \tilde{H}_k)^T + K_k R K_k^T. \end{aligned}$$

Here $\mathbf{Y}_k$ is the measured range and range rate, $G(\mathbf{X}_k^*, t_k)$ is the same pair computed from the reference, $\boldsymbol{\nu}_k$ is the innovation, $\tilde{H}_k$ is the 2×18 matrix of measurement partials, $R = \text{diag}(\sigma_\rho^2, \sigma_{\dot{\rho}}^2)$ is the measurement-noise covariance, K_k is the gain, and P_k^- , P_k^+ are the covariance before and after the update. The last equation is the Joseph form of the covariance update.

The reference trajectory update is applied between passes. After the filter processes the full arc, the deviation at the final time is mapped back to the epoch and added to the reference,

$$\mathbf{X}^*(t_0) \leftarrow \mathbf{X}^*(t_0) + \hat{\mathbf{x}}(t_0),$$

and the filter is rerun about the new reference until the pre-fit RMS stops dropping. For this data set that took three passes. The reference is held fixed within each pass.

Interpretation. I chose the Joseph form because it keeps the covariance symmetric and is less sensitive to round off than the plain update $P_k^+ = (I - K_k \tilde{H}_k) P_k^-$. That matters because μ , J_2 , and

C_D differ in size by about twenty orders of magnitude, so the covariance is badly scaled. Unlike the EKF, this filter does not relinearize at every measurement, so it does not get thrown off by a poor initial guess. The EKF was fragile on the drag coefficient and could leave residuals above the noise floor. Unlike the Potter square root filter, it needs no square root factor bookkeeping, which is not necessary here since the conditioning is manageable.

(b) Implementation and state estimation

The individualized observation file was processed over the full data arc (385 observations from $t = 0$ to 18340 s) using the MATLAB implementation of the chosen estimator. The estimated initial state and solve-for parameters are summarized below.

Estimated initial ECI state

Quantity	i	j	k
Position estimate (m)	757700.3073	5222606.5616	4851499.7554
Velocity estimate (m/s)	2213.250609	4678.372733	-5371.314407
Posterior 1σ for position (m)	7.525×10^{-3}	1.179×10^{-2}	1.484×10^{-2}
Posterior 1σ for velocity (m/s)	8.639×10^{-6}	1.444×10^{-5}	1.024×10^{-5}

Estimated dynamic parameters

Parameter	Estimate	Posterior 1σ
μ (m^3/s^2)	$3.98600400 \times 10^{14}$	4.157×10^5
J_2	$1.08200032 \times 10^{-3}$	2.446×10^{-10}
C_D	2.188027	3.807×10^{-3}

Estimated station coordinates

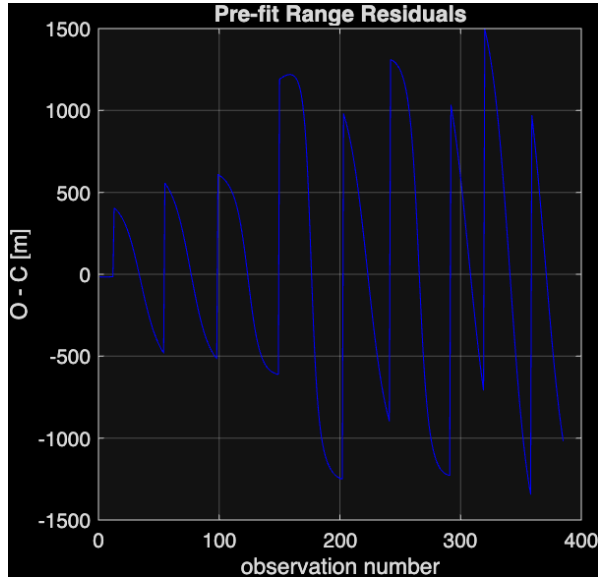
Station / quantity	X_s (m)	Y_s (m)	Z_s (m)
337 estimate	3860900.0030	3238499.9967	3898099.9752
337 posterior 1σ	5.271×10^{-3}	8.448×10^{-3}	8.761×10^{-3}
394 estimate	549500.0076	-1380869.9986	6182199.9776
394 posterior 1σ	7.339×10^{-3}	1.275×10^{-2}	1.656×10^{-2}

Interpretation. The estimated position and velocity stay very close to the nominal orbit, which says the a priori trajectory was already good. The parameters and station coordinates are pinned down by the data. μ and J_2 are determined to a relative level of about 10^{-9} and 10^{-7} . The drag coefficient C_D is the most weakly determined parameter at roughly 0.17%. The satellite position comes out at the millimeter level and the velocity at tens of micrometers per second. Station 394 was slightly less well determined than station 337.

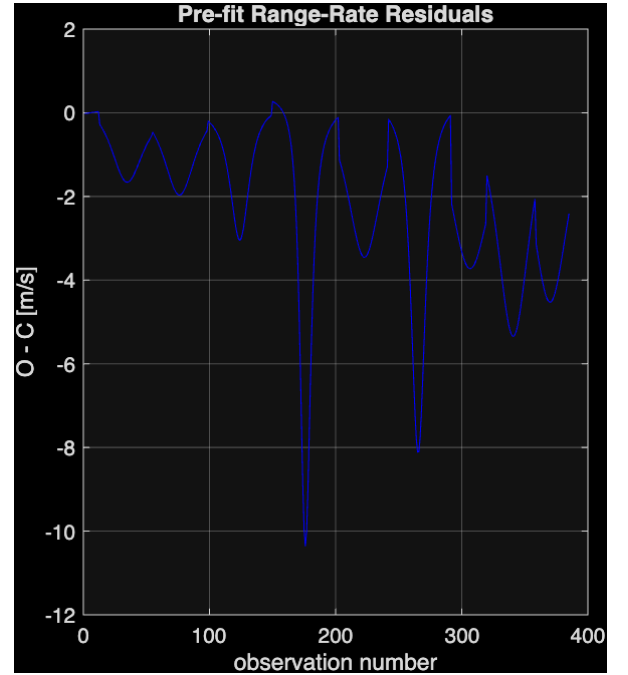
(c) Residual analysis

The pre-fit and post-fit residuals were computed for both range and range rate over the full observation arc. The corresponding RMS values are listed below.

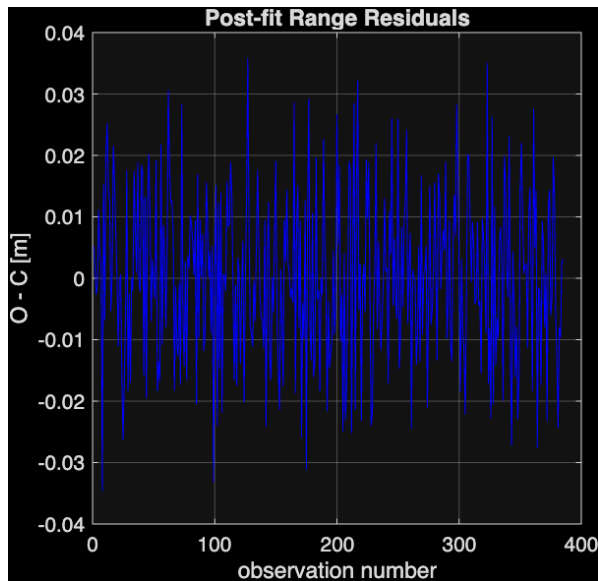
Residual metric	Pre-fit RMS	Post-fit RMS
Range (m)	7.328×10^2	1.352×10^{-2}
Range-rate (m/s)	2.900	1.432×10^{-3}



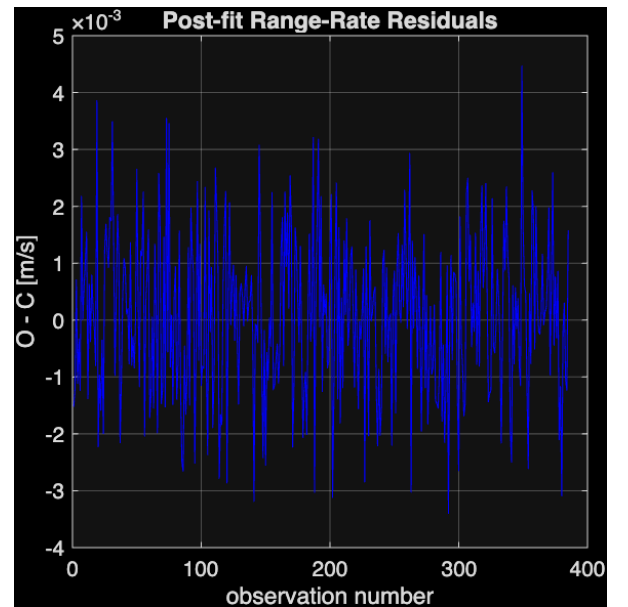
(a) Pre-fit range



(b) Pre-fit range rate



(c) Post-fit range



(d) Post-fit range rate

Figure 1: Pre-fit and post-fit residuals for range and range rate over the full observation arc.

Interpretation. The post-fit residuals are consistent with the stated observation noise levels of $\sigma_\rho = 0.01$ m and $\sigma_{\dot{\rho}} = 0.001$ m/s. Although their RMS comes out about 1.35 and 1.43 times those values. They have zero mean, no correlation from one point to the next, and no clustering by station pass, so there is no sign of bias or leftover modeling or linearization error. The small excess over the nominal noise means the actual noise in this data set is a little larger than the stated values. If R were scaled up by about $(1.35)^2$ in range and $(1.43)^2$ in range rate, the normalized residuals would come out near one. So, the sigmas would grow by the same factor without changing the estimate.

(d) Covariance and geometric interpretation

The final covariance was used to construct the position error ellipsoid associated with the satellite position estimate. Its semi-axes are 3.83, 6.66, and 18.89 mm, with a condition number of 24.3.

Interpretation. The ellipsoid indicates that the position uncertainty is strongly anisotropic, with the longest axis about 4.9 times the shortest. The longest principal axis suggests that the satellite position is least well determined in the along track direction. The shortest axis lies in the radial direction, which is the most strongly observed.

(e) Discussion

(a) Why is one station fixed in the filter formulation?

Range and range rate only depend on the relative geometry between the satellite and the stations. If the satellite state and all three station positions were solved for at once, there would be a direction the data cannot see. The whole satellite plus stations configuration could be shifted, or rotated, without changing any range or range rate value. That makes the information matrix rank deficient, so it cannot be inverted to get a covariance. Fixing one station removes that ambiguity. It ties the solution to a known point and sets the origin and orientation of the frame. This makes the rest of the state solvable. Station 101 is the one to fix because it is a fixed ground site. A moving platform would not give a stable reference.

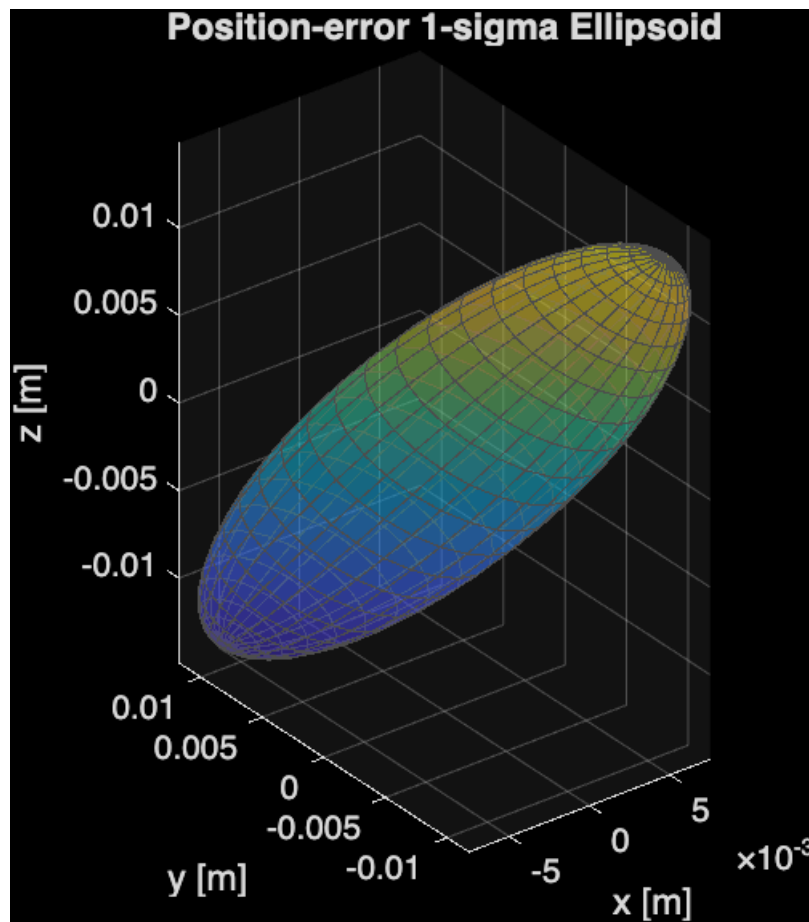


Figure 2: Position-error ellipsoid constructed from the final position covariance submatrix.

- (b) **Would the same result be obtained by omitting one station from the solve-for list entirely, rather than fixing it with near-zero covariance?**

Here station 101 is fixed by giving it a tiny a priori variance (10^{-10} m^2), rather than dropping it from the solve for list. In the limit where that variance goes to zero, the two approaches are equivalent for everything else. A station held with zero uncertainty does not move during the updates, and its correlations with the other parameters vanish. This means that the remaining estimates and their covariances come out the same as if the station had been left out of the state. Omitting the station treats it as a known, removes three parameters, and has no conditioning issue. Fixing it with a tiny variance keeps it in the state and puts a large entry into the a priori information matrix, which can negatively affect conditioning but is easy to loosen later. So the same result is obtained in the ideal limit. The choice comes down to conditioning and convenience.

- (c) **How would changing the a priori covariance and/or the observation-noise covariance be expected to affect the solution?**

The a priori covariance $\bar{P}_0$ controls how hard the solution is pulled toward the initial guess. A looser a priori lets the data take over and moves the estimate toward the data only answer. A tighter one holds it near the starting values. For well observed quantities it does not matter, because the measurements dominate. This is why the satellite state and the well tracked stations are insensitive to it. It matters more for weakly observed quantities and the fixed station is the extreme case. The observation noise covariance R controls how much the filter trusts

each measurement. A larger R gives a smaller gain, slower corrections, and a larger posterior covariance. A smaller R gives a larger gain, harder fitting, and a smaller posterior covariance. The relative size of the range and range rate entries sets how they trade off. This connects back to part (c), where the post fit residuals came out about 1.35 to 1.43 times the assumed σ . This means R is optimistic compared to the real noise. Scaling R up to match would raise the reported σ by about the same factor and bring the normalized residuals back to one, without changing the estimate.

- (d) **What numerical or convergence advantages does your chosen estimator offer relative to the other two listed choices?**

The Joseph covariance update stays symmetric on its own and is much less sensitive to round off than the plain short form update. So the covariance stays well even though the information matrix is badly scaled by the spread in parameter sizes. Together with relinearizing the reference between passes, it converged to the same answer as the batch in three passes with no tuning. The EKF, in contrast, updates the reference at every step, which is good for strong nonlinearity in a single pass but makes it fragile when the starting guess is far off." Same content, no contradiction. On this problem the early updates about a bad reference, with the weakly observed drag coefficient, tend to throw it off. Unless a warm up period is used, and even then it leaves residuals above the noise floor. Compared to the Potter square root filter, it is simpler to code. Potter carries a square root factor W with $P = WW^T$. This keeps the covariance positive definite and roughly doubles the working precision. This is the most robust choice for a severely ill conditioned problem, but at the cost of more bookkeeping and processing observations one component at a time. This problem is not conditioned badly enough to need that, so the Joseph form gives a stable, symmetric, positive covariance and clean convergence without the extra work.

Overall conclusion. The Joseph stabilized sequential filter converged in three passes to the same solution as the batch, with no tuning. It fit the observations down to the noise floor, with post-fit residuals that are noise like and only slightly above the stated noise levels. The final position uncertainty is at the millimeter level and strongly anisotropic, least well determined in the along-track direction and best in the radial direction, and the drag coefficient is the most weakly determined of the estimated parameters.
